

Customer Churn Prediction & Retention Intelligence

Project Type: Supervised Machine Learning Classification

Status: Final — Stakeholder-Ready Demonstration Report

Dataset: IBM Telco Customer Churn dataset

Classification Target: Customer churn (Yes / No)

Analysis Period: Simulated historical customer portfolio

Prepared for: Executive, Marketing, Customer Success, and Data Science Stakeholders

Important: All model results, metrics, financial estimates, and business outcomes in this report are **fictional but technically plausible** and are provided as a complete project example. They should not be represented as results from an actual production experiment.

1. Executive Summary

Customer churn represents a significant source of revenue loss for subscription-based businesses. The objective of this project was to develop a machine learning system capable of identifying customers who are likely to terminate their subscription, allowing the business to intervene before cancellation occurs.

A complete supervised classification workflow was designed around the IBM Telco Customer Churn dataset containing approximately 7,000 customer records and behavioral, contractual, demographic, and billing attributes.

Multiple supervised learning algorithms were evaluated, including:

- Logistic Regression
- K-Nearest Neighbors
- Support Vector Machine
- Decision Tree
- Random Forest
- Extra Trees
- Gradient Boosting
- HistGradientBoosting
- AdaBoost
- XGBoost

- LightGBM
- CatBoost

Rather than selecting a model based on a single train/test score, the models were subjected to preprocessing, hyperparameter tuning, stratified cross-validation, and out-of-sample evaluation.

The final model selected for this demonstration was **CatBoost**, which produced the strongest overall balance between discrimination, recall, precision, calibration, and operational usefulness.

Final validation performance

Metric	Final Model
ROC-AUC	0.846
PR-AUC	0.681
Accuracy	0.789
Precision	0.642
Recall	0.714
F1 Score	0.676
Specificity	0.824
Balanced Accuracy	0.769

The model's most important predictive signals were:

1. Contract type
2. Tenure
3. Monthly charges
4. Internet service type
5. Payment method
6. Total charges
7. Online security subscription

8. Technical support subscription

The analysis of model predictions indicates that churn is concentrated among customers with **short tenure, month-to-month contracts, relatively high monthly charges, and limited additional services.**

The most important business conclusion is that churn is not distributed uniformly across the customer base. A relatively small, identifiable segment exhibits substantially elevated churn risk.

2. Business Problem

2.1 Problem Statement

The business currently reacts to customer cancellations after they occur. This creates two problems:

1. The opportunity to retain the customer has already been lost.
2. Retention campaigns cannot be efficiently targeted because customers are treated similarly.

The proposed machine learning solution answers:

Which currently active customers are most likely to churn, and what characteristics distinguish high-risk customers from low-risk customers?

The output is not intended to replace business decisions. Instead, it provides a ranked list of customers that can be used by customer-success and marketing teams.

2.2 Business Objective

The project has four objectives:

Primary objective

Predict whether an active customer will churn.

Secondary objectives

- Rank customers by churn probability.
- Identify the behavioral characteristics associated with churn.

- Estimate the size of the high-risk customer population.
- Provide actionable retention recommendations.

3. Dataset

The IBM Telco Customer Churn dataset contains approximately 7,000 records describing telecommunications customers.

Each record represents one customer.

The target variable is:

Churn

with:

Yes = customer churned

No = customer remained

Major feature groups

Category	Examples
Demographics	Gender, Senior Citizen, Partner, Dependents
Account	Tenure, Contract, Payment Method
Billing	Monthly Charges, Total Charges
Internet	Internet Service
Services	Online Security, Backup, Device Protection
Support	Tech Support
Communication	Streaming TV, Streaming Movies
Administrative	Paperless Billing

4. Data Preparation

4.1 Initial Data Audit

The following data-quality checks were performed:

- Duplicate records
- Missing values
- Invalid numeric values
- Inconsistent categorical values
- Cardinality analysis
- Target imbalance
- Outlier inspection
- Distribution analysis

A particularly important issue was the TotalCharges variable, which contained values stored as strings and several records that could not be interpreted directly as numerical values.

These observations were converted to missing values and subsequently handled through the preprocessing pipeline.

5. Exploratory Data Analysis

5.1 Target Distribution

The target variable was moderately imbalanced.

Approximately:

Churn Status Share

Stayed	73.5%
Churned	26.5%

This means that a naïve model predicting **"No Churn" for every customer** would achieve approximately 73.5% accuracy while being completely useless for retention.

Therefore, accuracy was explicitly prevented from becoming the sole model-selection criterion.

6. Important Relationships Discovered During EDA

6.1 Contract Type

Contract type showed one of the strongest relationships with churn.

Illustrative churn rates:

Contract	Churn Rate
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Month-to-month	42.1%
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One year	11.3%
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Two year	2.8%
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This is a major business finding.

Customers without long-term contractual commitment are dramatically more likely to leave.

6.2 Tenure

Tenure was strongly negatively associated with churn.

Illustrative results:

Tenure	Churn Rate
--------	------------

0–6 months	48.7%
------------	--------------

7–12 months	36.4%
-------------	-------

13–24 months	25.1%
--------------	-------

25–48 months	17.2%
--------------	-------

49+ months	8.4%
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The relationship suggests that the earliest stage of the customer lifecycle is particularly important.

This indicates that retention efforts should not be limited to customers who have already demonstrated dissatisfaction.

6.3 Monthly Charges

Customers paying higher monthly charges demonstrated elevated churn.

Illustrative relationship:

Monthly Charge Segment Churn Rate

Low	15.8%
Medium	23.4%
High	37.9%

However, high monthly charges alone do not explain churn.

The interaction between **high monthly charges + short tenure + month-to-month contract** is substantially more concerning.

6.4 Payment Method

Electronic check customers showed a materially higher churn rate than customers using automatic payment methods.

Illustrative churn rates:

Payment Method Churn Rate

Electronic check	44.6%
Mailed check	19.2%
Bank transfer	17.5%
Credit card	15.9%

This finding suggests that payment friction or customer characteristics associated with payment method may contribute to churn.

It should not automatically be interpreted as a causal effect.

7. Feature Engineering

Several transformations were introduced to improve model performance.

Tenure groups

Tenure was additionally represented through lifecycle categories:

New

Early

Established

Loyal

Charge-to-tenure relationships

Derived variables were introduced to represent customer value and billing characteristics.

Examples:

AverageMonthlyValue

ServiceCount

HasSupportService

HasSecurityService

ContractTenureInteraction

Service count

The number of optional services subscribed to was aggregated into a single feature.

This allowed the model to capture differences between:

Basic customers

and

Highly integrated customers

8. Machine Learning Methodology

The dataset was separated into:

80% Training

20% Holdout Test

The test set was isolated before model development and was never used for hyperparameter optimization.

The training set was evaluated using:

5-fold stratified cross-validation

Stratification was used to preserve the churn/non-churn distribution in every fold.

9. Candidate Models

The following supervised classification algorithms were evaluated.

Linear models

- Logistic Regression

Distance-based models

- K-Nearest Neighbors

Kernel methods

- Support Vector Machine

Tree-based models

- Decision Tree
- Random Forest
- Extra Trees

Boosting methods

- AdaBoost
- Gradient Boosting
- HistGradientBoosting
- XGBoost
- LightGBM
- CatBoost

This provides a broad comparison across fundamentally different model families rather than assuming that one algorithm is universally optimal.

10. Hyperparameter Optimization

Hyperparameters were optimized using randomized or Bayesian-style search depending on the model family.

Examples included:

Logistic Regression

C

penalty

class_weight

solver

Random Forest

n_estimators

max_depth

min_samples_split

min_samples_leaf

max_features

class_weight

SVM

C

gamma

kernel

class_weight

XGBoost / LightGBM / CatBoost

learning_rate

n_estimators

max_depth

subsample
colsample
regularization
min_child_samples
l2_regularization

The optimization objective was primarily **ROC-AUC**, with additional evaluation using PR-AUC, recall, precision, and F1.

11. Cross-Validation Results

The following table summarizes the fictional experimental results.

Model	CV ROC-AUC	CV F1	CV Recall	CV Precision
Logistic Regression	0.831	0.650	0.704	0.604
KNN	0.759	0.566	0.611	0.528
Decision Tree	0.742	0.545	0.583	0.514
Random Forest	0.821	0.626	0.669	0.589
Extra Trees	0.816	0.617	0.658	0.581
SVM	0.826	0.642	0.697	0.596
AdaBoost	0.816	0.618	0.674	0.571
Gradient Boosting	0.838	0.654	0.699	0.616
HistGradientBoosting	0.841	0.661	0.706	0.623
XGBoost	0.844	0.667	0.710	0.630
LightGBM	0.843	0.665	0.706	0.629
CatBoost	0.848	0.674	0.716	0.636

12. Model Selection

CatBoost achieved the highest cross-validated ROC-AUC and F1 score while also producing strong recall.

The model was therefore selected as the champion model.

The decision was not based solely on the highest numerical score.

The final model had a favorable combination of:

- discrimination
- recall
- precision
- robustness
- categorical-feature handling
- relatively low preprocessing complexity
- interpretability through feature importance and SHAP analysis

13. Final Holdout Evaluation

After selecting the model using the training data and cross-validation, the final model was evaluated once on the untouched test set.

Holdout results

Metric	Result
ROC-AUC	0.846
PR-AUC	0.681
Accuracy	0.789
Precision	0.642
Recall	0.714
F1	0.676
Specificity	0.824

Metric	Result
--------	--------

Balanced Accuracy	0.769
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The model therefore identifies approximately **71% of customers who eventually churn** at the selected operating threshold.

14. Confusion Matrix

Illustrative holdout confusion matrix:

	Predicted Stay	Predicted Churn
Actual Stay	1,038	222
Actual Churn	150	375

Interpretation:

- **375** churners were correctly identified.
- **150** churners were missed.
- **222** customers were incorrectly flagged as churn risks.
- **1,038** customers were correctly classified as non-churners.

The model therefore trades some false positives for substantially better churn detection.

This is appropriate for a retention use case because missing a customer who is about to churn can be more expensive than contacting a customer who ultimately stays.

15. ROC Curve Analysis

The ROC-AUC of **0.846** indicates strong ranking ability.

A random classifier would have:

ROC-AUC = 0.50

while:

ROC-AUC = 1.00

would represent perfect discrimination.

A value of 0.846 therefore indicates that the model can effectively distinguish between higher- and lower-risk customers.

However, ROC-AUC should not be interpreted as saying that 84.6% of predictions are correct.

It measures ranking/discrimination performance across thresholds.

16. Precision-Recall Analysis

Because only approximately one-quarter of customers churn, precision-recall analysis is particularly important.

The model achieved:

PR-AUC = 0.681

This indicates substantially better performance than random targeting.

For a retention team, this means that prioritizing customers based on predicted probability produces a much more concentrated pool of likely churners than contacting customers randomly.

17. Threshold Optimization

The default classification threshold of 0.50 was not automatically considered optimal.

Different business strategies require different thresholds.

Example operating points

Threshold Precision Recall Business Interpretation

0.30	0.52	0.86	Aggressive retention
0.40	0.59	0.79	High-recall strategy
0.50	0.64	0.71	Balanced strategy
0.60	0.71	0.61	More selective
0.70	0.78	0.47	Premium targeting

The recommended default operating point is **0.50** for general retention.

However, the business should ultimately select the threshold using expected financial value rather than a generic machine-learning metric.

18. Feature Importance

The model identified the following features as the strongest predictive signals.

Rank	Feature	Relative Importance
1	Contract	18.4%
2	Tenure	15.7%
3	Monthly Charges	12.9%
4	Internet Service	9.4%
5	Payment Method	8.2%
6	Total Charges	7.1%
7	Online Security	6.5%
8	Tech Support	5.9%
9	Service Count	4.8%
10	Paperless Billing	3.9%

Feature importance describes predictive contribution, not causality.

For example, the model identifying contract type as important does not prove that changing a customer's contract will cause that customer to stay.

19. SHAP-Based Interpretation

SHAP analysis was used to understand how individual features affected predictions.

The general patterns were:

Increasing churn probability

- Month-to-month contract

- Short tenure
- High monthly charges
- Electronic check payment
- Fiber internet
- No technical support
- No online security
- Few subscribed services

Decreasing churn probability

- Two-year contract
- Long tenure
- Automatic payment
- Multiple subscribed services
- Technical support
- Online security
- Lower relative monthly charges

This allows the model to move beyond:

"Customer has a 78% probability of churn."

and toward:

"Customer has a 78% predicted churn probability, primarily driven by short tenure, month-to-month contract, high monthly charges, and lack of support services."

That distinction is important for operational adoption.

20. Customer Risk Segmentation

Predicted probabilities were converted into operational risk groups.

Risk Segment Probability Customer Share Illustrative Churn Rate

Very Low	<20%	35%	7%
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Risk Segment	Probability	Customer Share	Illustrative Churn Rate
Low	20–40%	25%	21%
Medium	40–60%	18%	43%
High	60–80%	14%	68%
Critical	>80%	8%	86%

Low	20–40%	25%	21%
Medium	40–60%	18%	43%
High	60–80%	14%	68%
Critical	>80%	8%	86%

The highest-risk 8% of customers therefore represents a disproportionately valuable intervention population.

21. Analysis of Predicted Customers

The most useful output of the project is not the overall accuracy score.

It is the analysis of **which customers the model predicts will churn**.

The high-risk population displayed a very different profile from the low-risk population.

High-risk customers were more likely to:

- have month-to-month contracts
- have short tenure
- pay relatively high monthly charges
- use electronic check payments
- lack technical support
- lack online security
- use fiber internet
- subscribe to fewer additional services

This provides a direct bridge between machine learning and business strategy.

22. High-Risk Customer Archetype

The model effectively identified a recurring customer archetype:

New, month-to-month customer paying relatively high monthly fees with limited additional services and no technical-support subscription.

This customer should be considered a priority for proactive retention.

A hypothetical customer matching this profile might receive:

Predicted churn probability: 82%

Risk: Critical

Primary drivers:

- Month-to-month contract
- Tenure < 12 months
- High monthly charge
- No technical support
- No online security

The prediction is therefore actionable rather than merely descriptive.

23. Important Interaction Effects

The analysis indicates that individual variables do not tell the complete story.

For example:

Month-to-month + short tenure

produces considerably greater risk than either characteristic individually.

Similarly:

High monthly charge + limited services

creates a potentially unfavorable perceived value proposition.

This suggests that customers may not simply be "price sensitive."

They may be evaluating:

Perceived value

=

Services received

/

Monthly cost

Customers paying high prices while receiving relatively few services may therefore represent an important retention segment.

24. Customer Lifecycle Insight

One of the strongest insights from the analysis is that churn risk is heavily concentrated during the early customer lifecycle.

This suggests that retention should be treated as a **lifecycle problem**, not merely a cancellation problem.

A possible intervention strategy is:

Days 0–30

Onboarding and product education.

Days 30–90

Usage monitoring and support engagement.

Months 3–6

Identify dissatisfaction and service gaps.

Months 6–12

Promote appropriate long-term plans or bundles.

After 12 months

Focus increasingly on loyalty and customer lifetime value.

25. Recommended Business Actions

Action 1 — Prioritize Critical-Risk Customers

Customers with:

Predicted churn probability > 80%

should receive immediate review.

Possible interventions:

- personalized retention offer
 - support outreach
 - plan optimization
 - service-bundle recommendation
 - contract review
-

Action 2 — Improve Early-Customer Retention

New customers should receive additional attention during their first six months.

Recommended interventions:

- onboarding campaigns
 - proactive technical assistance
 - product education
 - satisfaction surveys
 - usage-based recommendations
-

Action 3 — Investigate Month-to-Month Customers

Month-to-month customers represent a major risk segment.

Instead of indiscriminately offering discounts, the company should investigate:

- price sensitivity
 - service dissatisfaction
 - competitor switching
 - technical problems
 - contract flexibility
-

Action 4 — Address Perceived Value

High monthly charges become particularly concerning when customers use relatively few additional services.

The business could test:

- bundled packages
 - personalized plan recommendations
 - loyalty discounts
 - service upgrades
 - transparent pricing
-

Action 5 — Increase Payment Automation

The strong relationship between electronic-check customers and churn suggests that payment experience deserves further investigation.

Possible experiments:

- automatic-payment incentives
- simplified payment setup
- payment reminders
- payment-method education

This should be tested experimentally before claiming that payment method itself causes churn.

26. Hypothetical Financial Impact

Assume the business has:

100,000 active customers

and an annual churn rate of:

25%

This implies:

25,000 expected annual churners

Assume:

Average annual customer value = \$600

The theoretical annual revenue exposed to churn would be:

$25,000 \times \$600$

$= \$15,000,000$

Suppose the model identifies 70% of potential churners and the retention program successfully saves 20% of those customers.

Then:

$25,000 \times 70\% \times 20\%$

$= 3,500$ retained customers

At \$600 annual value:

$3,500 \times \$600$

$= \$2.1\text{M}$

This produces a theoretical gross value of approximately:

\$2.1 million annually.

This is **not a forecast of actual financial performance**. It is an illustration of how model performance can be translated into an economic framework.

27. Cost-Sensitive Decision Framework

The business should ultimately select the prediction threshold using:

Expected Value

=

True Positives $\times$ Retention Benefit

-

False Positives $\times$ Intervention Cost

-

Campaign Costs

For example, if:

Retention value = \$600

Retention probability after intervention = 20%

Intervention cost = \$30

then the expected benefit of contacting a genuinely high-risk customer can substantially exceed the intervention cost.

This economic formulation is preferable to selecting a threshold solely because it maximizes F1.

28. Model Explainability

A retention model must be explainable to business stakeholders.

For each prediction, the production system should ideally provide:

Customer ID

Predicted churn probability

Risk segment

Top 3–5 contributing factors

Recommended action

Prediction timestamp

Model version

Example:

Field	Example
Customer	CUST-10492
Churn Probability	82%
Risk	Critical

Field	Example
Driver 1	Month-to-month contract
Driver 2	Short tenure
Driver 3	High monthly charge
Driver 4	No technical support
Recommended Action Proactive retention outreach	

29. Model Validation Beyond Accuracy

A production deployment should not stop at offline performance.

The model should be evaluated for:

Calibration

A group of customers predicted at 70% risk should actually churn at approximately 70%, subject to sampling uncertainty.

Stability

Performance should remain reasonable across different customer cohorts and time periods.

Fairness

Performance should be monitored across relevant demographic groups.

Drift

Changes in:

- customer behavior
- pricing
- products
- contracts
- competitors
- payment systems

may change the relationship between features and churn.

30. Production Architecture

A practical implementation could follow:

CRM / Billing Database

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v

Data Pipeline

|

v

Feature Engineering

|

v

ML Model

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v

Churn Probability Scores

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v

Risk Segmentation

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|

|

v

v

Customer Success Marketing

|

|

+-----+-----+

|

v

Retention Campaign

|

v

New Outcomes

|

v

Model Monitoring

The model should be retrained periodically using newly observed churn outcomes.

31. Monitoring Strategy

The production system should monitor three categories.

Data Monitoring

Track:

- missing-value rates
- category distribution
- feature ranges
- unexpected values
- data freshness

Model Monitoring

Track:

- ROC-AUC
- PR-AUC
- recall

- precision
- calibration
- prediction distribution

Business Monitoring

Track:

- retention rate
 - campaign conversion
 - intervention cost
 - incremental retained revenue
 - customer lifetime value
 - false-positive intervention rate
-

32. Recommended A/B Test

The strongest way to establish business value is not simply to observe that the model predicts churn.

The retention intervention should be experimentally evaluated.

Customers identified as high-risk should be randomly divided into:

Treatment Group

and

Control Group

The treatment group receives the retention intervention.

The control group receives the normal customer experience.

The key metric becomes:

Incremental Retention Rate

rather than simply:

Model Accuracy

This distinguishes:

predictive power

from:

economic impact.

33. Limitations

Several limitations must be recognized.

1. Correlation is not causation

The model identifies characteristics associated with churn. It does not prove that changing those characteristics will prevent churn.

2. Historical bias

Historical customer behavior may contain biases that the model reproduces.

3. Dataset limitations

The dataset does not contain every variable that may influence churn.

For example:

- competitor activity
- customer sentiment
- service outages
- call-center interactions
- geographic market conditions
- competitor pricing

may materially influence churn.

4. Temporal leakage risk

A production implementation must ensure that features are calculated only from information available **before the prediction date**.

5. Concept drift

Customer behavior can change over time.

A model trained today may degrade as products and market conditions change.

34. Final Model Assessment

The CatBoost classifier demonstrated the strongest overall performance among the evaluated candidate models.

Its key validation characteristics were:

ROC-AUC 0.846

PR-AUC 0.681

Recall 0.714

Precision 0.642

F1 0.676

More importantly, the model produces a useful ranking of customers by risk.

The resulting predictions reveal a coherent business pattern rather than a collection of statistically opaque classifications.

35. Key Findings for Executives

Finding 1

Churn is concentrated rather than uniformly distributed.

The highest-risk customer segment represents a disproportionately large share of expected churn.

Finding 2

The first months of the customer lifecycle are critical.

Short-tenure customers have substantially greater churn risk.

Finding 3

Contract structure is one of the strongest predictive signals.

Month-to-month customers are significantly more vulnerable than customers with long-term contracts.

Finding 4

Price and perceived value appear to interact.

High monthly charges become particularly concerning when customers have relatively few additional services.

Finding 5

Support and security services are associated with lower churn.

Customers with deeper service adoption exhibit substantially lower predicted risk.

Finding 6

Targeted intervention is more efficient than universal retention campaigns.

The model allows the business to concentrate resources on customers with the highest expected value from intervention.

36. Final Recommendation

The proposed churn prediction system is suitable as a **decision-support system**, subject to further validation using real production data.

The recommended strategy is:

1. Deploy the model initially in **shadow mode**.
2. Generate churn probabilities without automatically contacting customers.
3. Validate calibration and data quality.
4. Select the operating threshold using financial cost-benefit analysis.
5. Launch a controlled retention experiment.
6. Measure incremental retention rather than model accuracy alone.
7. Monitor model drift and business performance.
8. Retrain periodically as new customer outcomes become available.

The central strategic recommendation is:

Move from reactive churn management to proactive, risk-based customer retention.

The machine learning model provides the prioritization mechanism; customer-success interventions and controlled experimentation determine whether that prioritization produces measurable financial value.

Appendix A — Example End-to-End ML Pipeline

Raw Customer Data

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Data Validation

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Exploratory Data Analysis

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Train/Test Split

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Preprocessing

↓

Feature Engineering

↓

Stratified 5-Fold CV

↓

Hyperparameter Optimization

↓

Candidate Model Comparison

↓

Champion Model Selection

↓

Final Training

↓

Untouched Test Evaluation



Probability Calibration



Threshold Optimization



Customer Risk Scoring



SHAP / Explainability



Business Segmentation



Retention Experiment



Production Monitoring

Appendix B — Recommended Model Selection Criteria

A production model should not be selected from one metric.

The recommended hierarchy is:

1. **Business value**
2. **Out-of-sample performance**
3. **Recall/precision trade-off**
4. **Probability calibration**
5. **Model stability**
6. **Interpretability**
7. **Computational cost**
8. **Operational complexity**

For churn prediction, a model with marginally lower ROC-AUC but significantly better calibration or business value may be preferable to the theoretical leaderboard winner.

Appendix C — Final Project Deliverables

A complete implementation of this project would contain:

project/

|

| — data/

|

| | — raw/

|

| | — processed/

|

| — notebooks/

|

| | — 01_eda.ipynb

|

| | — 02_feature_engineering.ipynb

|

| | — 03_model_comparison.ipynb

|

| | — 04_model_interpretation.ipynb

|

| — src/

|

| | — preprocessing.py

|

| | — features.py

|

| | — train.py

|

| | — evaluate.py

|

| | — predict.py

|

| | — monitoring.py

|

| — models/

```
|  └─ churn_model.pkl
|
|  └─ reports/
|
|  └─ model_metrics.json
|  └─ feature_importance.csv
|  └─ stakeholder_report.pdf
|
└─ README.md
```

Conclusion

This project demonstrates a complete supervised classification workflow rather than simply fitting a classifier and reporting accuracy.

The workflow incorporates:

- business problem formulation
- data auditing
- exploratory analysis
- feature engineering
- multiple supervised learning algorithms
- hyperparameter tuning
- stratified cross-validation
- model comparison
- holdout evaluation
- threshold optimization
- confusion-matrix analysis
- ROC-AUC and PR-AUC
- feature importance
- SHAP-based interpretation

- customer segmentation
- financial reasoning
- deployment architecture
- monitoring
- experimentation
- business recommendations

The most important outcome is therefore not the reported **0.846 ROC-AUC**.

The important outcome is the transformation of raw customer records into a **ranked, explainable, economically actionable retention strategy**.